Franck Portier
2016-2017

M1 Macro

Group A

Session 10

These notes are written during class. They are posted "for the record" and may content mistakes. They are not intented to be read by anyone that did not attend the class.

$$\tilde{y}_t = \rho \tilde{y}_{t-1} + \underbrace{\varepsilon^y_t}$$

$$\underset{t}{E}(\varepsilon^y_t) = 0 \ , \ iid$$

Information set: everything up to period t

$$\underset{t}{E}(\tilde{y}_{t+1}) = \underset{t}{E}(\rho \tilde{y}_t + \varepsilon^y_{t+1}) = \rho \tilde{y}_t + \underbrace{\underset{t}{E}\left[\varepsilon^y_{t+1}\right]}_{0}$$

realization = a real number

$$\rightarrow E_t(\tilde{y}_{t+1}) = \rho \tilde{y}_t$$

$$E_t(\tilde{y}_{t+2}) = E_t(\rho \tilde{y}_{t+1} + \varepsilon^y_{t+2})$$

$$= E_t\left[\rho(\rho \tilde{y}_t + \varepsilon^y_{t+1}) + \varepsilon^y_{t+2}\right]$$

$$= E_t\left[\rho^2 \tilde{y}_t + \rho\, \varepsilon^y_{t+1} + \varepsilon^y_{t+2}\right]$$

$$= \rho^2 \tilde{y}_t + \rho\, \underbrace{E_t\left[\varepsilon^y_{t+1}\right]}_{0} + \underbrace{E_t\left[\varepsilon^y_{t+2}\right]}_{0}$$

$$= \rho^2 \tilde{y}_t$$

$$E_t\left[\tilde{y}_{t+n}\right] = \rho^n \tilde{y}_t$$

Euler eq.

$$/A_{t+1} : \quad E_t\left[-u'(c_t) + \beta(1+r)\, u'(c_{t+1})\right] = 0$$

$$\underbrace{E_t\left[u'(c_t)\right]} = E_t\left[\beta(1+r) \times u'(c_{t+1})\right]$$

$$u'(c_t) = \beta(1+r)\, E_t\left[u'(c_{t+1})\right]$$

$$\boxed{\beta(1+r) = 1} \qquad \boxed{u'(c_t) = E_t\, u'(c_{t+1})}$$

Assume $u(c_t) = \alpha_0 c_t - \frac{\alpha_1}{2} c_t^2$

$$u'(c_t) = \alpha_0 - \alpha_1 c_t$$

Plug into the Euler equation:

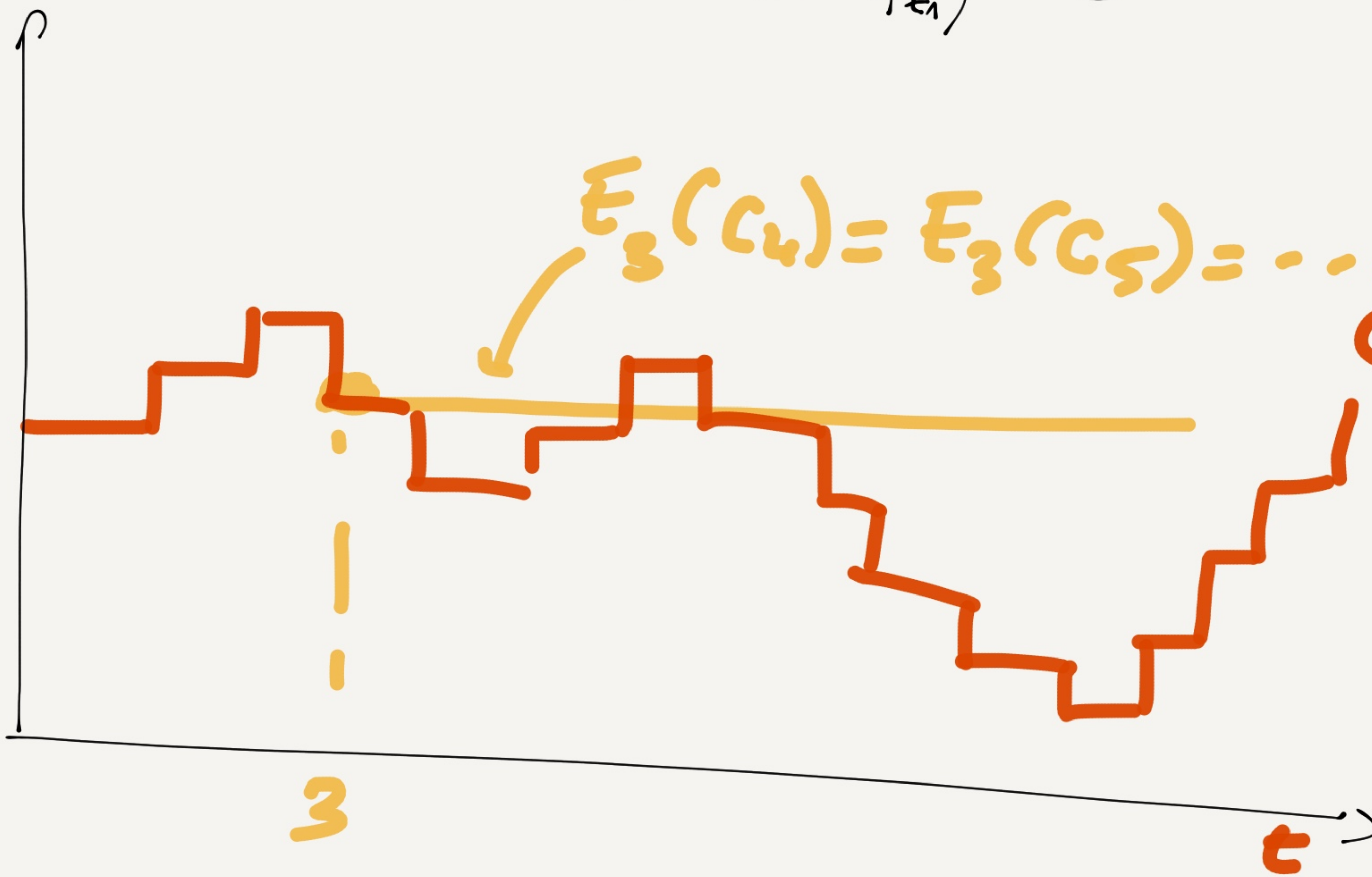
$$u'(c_t) = E_t\, u'(c_{t+1})$$

$$\alpha_0 - \alpha_1 c_t = E_t\left[\alpha_0 - \alpha_1 c_{t+1}\right]$$

$$\Rightarrow \quad \boxed{c_t = E_t\left[c_{t+1}\right]}$$

$$E_t C_{t+1} = C_t \iff E_t(C_{t+1} - C_t) = 0$$

$$E_t(\Delta C_{t+1}) = 0$$



Process s.t. $\quad E_t x_{t+1} = x_t$

$\Rightarrow \quad x_{t+1} = x_t + \varepsilon_{t+1}, \quad \varepsilon_{t+1}$: iid, $E(\varepsilon_{t+1}) = 0$

check $\quad E_t(x_{t+1}) = E_t(x_t + \varepsilon_{t+1}) = x_t$